

## Phase 5 implication decisions

| claim                                                    | decision    | evidence                                                               |
|----------------------------------------------------------|-------------|------------------------------------------------------------------------|
| exact lossless affine representation                     | supported   | machine-precision reconstruction and proof                             |
| additive phase winds                                     | refuted     | phase domain has length $\pi$                                          |
| Cayley is the original additive process                  | refuted     | it is an invertible nonlinear transform                                |
| Cayley is circular Brownian motion                       | refuted     | angular diffusion is state dependent                                   |
| one driver creates rank-one terminal law in every system | refuted     | hypoelliptic control has full-rank covariance                          |
| complex embedding creates holomorphic derivatives        | refuted     | normal derivative depends on off-curve extension                       |
| phase coordinate reduces ordinary MC variance            | refuted     | identical streams give identical samples                               |
| phase coordinate lowers direct terminal-estimator work   | refuted     | same random dimension plus an arctangent/tangent round trip            |
| compactification always improves PDE numerics            | refuted     | accuracy and conditioning depend on $c$ and resolution                 |
| compatible multiplicative phase can wind losslessly      | supported   | logarithmic spiral reconstruction                                      |
| geometry can preserve a transformed invariant            | supported   | exact Cayley map preserves unit modulus                                |
| new analytic solution for ABM                            | not claimed | exact phase kernel is the known Gaussian kernel under a diffeomorphism |